

OSTEOARTHRITIS CASE ASSIGNMENT

PATIENT INFORMATION

Name	Aydan Güler	Age / Sex	65 years / Female
Height / Weight	1.60 m / 85 kg	BMI	33.2 kg/m ² (Obese Class I)
Chief Complaint	Right knee pain, worst in the morning	Diagnosis	Degenerative Knee Osteoarthritis

1. IDENTIFICATION OF THE PROBLEM (DIAGNOSIS)

The patient is a 65-year-old obese female presenting with right knee pain, particularly pronounced in the morning. Physical examination and clinical findings are consistent with degenerative knee osteoarthritis (OA). The diagnosis has been explained to the patient.

- Osteoarthritis is a degenerative joint disease caused by progressive loss of articular cartilage.
- It is the most common form of arthritis and a leading cause of disability in elderly patients.
- Risk factors present in this patient: age (>60), female sex, obesity (BMI 33.2), and likely physical inactivity.
- The disease may be asymptomatic early on, but if left untreated it leads to progressive joint damage, loss of function, and reduced quality of life.
- Symptoms include joint pain (worse with activity and in the morning), stiffness, crepitus, and limited range of motion.

2. THE PURPOSE OF TREATMENT

- Relieve pain and reduce morning stiffness to improve daily functioning.
- Slow disease progression and preserve remaining cartilage and joint structure.
- Reduce body weight to decrease mechanical load on the knee joint.
- Prevent complications: functional disability, falls, loss of independence.
- Educate the patient and improve quality of life.

3. NON-DRUG TREATMENT

Weight Management (Priority)

- This patient is obese (BMI 33.2). Each 1 kg of weight loss reduces knee joint load by ~4 kg per step.
- Target: 5–10% body weight reduction (approx. 4–8 kg).
- Low-calorie diet: avoid pasta, rice, sugary foods, pastries. Increase vegetables, fruits, and fiber.
- Reduce solid (saturated) fats; prefer liquid fats (olive oil). Avoid red meat; prefer chicken, fish, legumes.

- Referral to a dietitian is recommended.

Exercise

- Regular low-impact aerobic exercise: walking 30–45 minutes, at least 3 days per week.
- Quadriceps-strengthening exercises to improve knee joint stability and reduce load.
- Swimming and cycling are particularly suitable — minimal joint impact.
- Avoid high-impact activities: running, jumping, prolonged stair climbing.

Lifestyle Modifications

- Salt restriction: cook meals without added salt to reduce fluid retention (relevant given obesity + future NSAID risk).
- Smoking history to be questioned; cessation strongly advised.
- Alcohol: if used, restriction or cessation recommended.
- Activity modification: reduce prolonged standing, arrange for seated rest during social visits.

Physical Therapy & Assistive Devices

- Referral to physiotherapy: TENS, therapeutic ultrasound, hot/cold packs for pain and stiffness.
- Walking cane (held in contralateral hand) to offload the affected right knee.
- Patient education: explain the chronic nature of OA, the importance of adherence, and realistic goals.

4. SUITABILITY OF THE SELECTED DRUG

Other Medications Currently Used

- Patient was questioned about current medication use.
- No concurrent medications reported. Relevant drug interactions: avoid combining paracetamol with warfarin or hepatotoxic drugs.

Comorbidities Affecting Drug Choice

- Obesity: increases GI, renal, and cardiovascular risk — long-term oral NSAIDs should be avoided.
- Age ≥ 65 : reduced renal clearance — start at lowest effective doses; monitor renal function.
- No history of peptic ulcer, renal failure, or cardiovascular disease reported.

Pregnancy / Breastfeeding

- Patient is 65 years old — not applicable.

Social Security

- Social security status confirmed. Prescribed medications are available under national insurance coverage.

Smoking / Alcohol Use

- Smoking and alcohol history questioned. Cessation/restriction advised as above.

5. THE CHOICE OF PHARMACOTHERAPY — PRESCRIPTION

Drug Selection Rationale

Priority	Drug	Dose	Reason
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1st line	Paracetamol 500 mg	1–2 tabs every 6–8h PO. Max 4 g/day.	Safest analgesic for elderly. Avoids GI/renal/CV risks of NSAIDs.
Add-on	Diclofenac 1% gel (topical)	Apply 2–4 g to knee 3–4x/day.	Local anti-inflammatory. Minimal systemic absorption.
2nd line*	Naproxen 250–500 mg + Omeprazole 20 mg	Naproxen BD PO. Omeprazole OD PO.	Only if paracetamol insufficient. Lowest CV risk NSAID. Always add PPI.
Flares	Intra-articular corticosteroid injection	Max 3–4 times/year.	For acute exacerbations. Short-acting local effect.

6. Comparison of Drug Groups (MAUA)

Drug	Efficacy	Safety	Suitability	Cost	Total
Paracetamol	4	9	9	10	8
Topical Diclofenac Gel	7	9	8	7	7.75
Ibuprofen	8	5	8	9	7.5
Naproxen	8	5	7	8	7
Celecoxib	8	7	8	5	7
Tramadol	6	4	6	6	5.5

PRESCRIPTION

Physician: LOGEN MOHAMED SAPIL Diploma no :20224507 Address:lefkoşa
 Tel no: 0537376806 Date: 6/6/2026. Patient: Aydan Güler Age: 65

1. Paracetamol 500 mg tablet

Take 1–2 tablets by mouth every 6–8 hours as needed for pain. Maximum 4 g/day.
 Quantity: 60 tablets. Use regularly for 4 weeks.

2. Diclofenac sodium 1% gel (e.g., Voltaren Emulgel 100 g tube)

Apply 2–4 g to the right knee 3–4 times daily. Rub in gently. Wash hands after use.
 Quantity: 1 tube (100 g). Duration: 4 weeks.

Instructions: Do not exceed maximum paracetamol dose. Avoid alcohol. Store at room temperature.

Signature: _____

7. DURATION OF TREATMENT

- Paracetamol: Use as needed for pain. Initial course: 4 weeks. Can be continued long-term given its safety profile.

- Topical Diclofenac gel: 4–6 weeks initially; can be continued as needed.
- Oral NSAIDs (if escalated): Shortest duration possible — no longer than 2–4 weeks continuously.
- Non-pharmacological measures (exercise, diet, physiotherapy): Lifelong — these are the most important long-term interventions.
- Intra-articular corticosteroid: As needed during flares; maximum 3–4 injections per year in the same joint.

8. INFORMATION, INSTRUCTIONS & WARNINGS TO THE PATIENT

Effect of the Drug

- Paracetamol works to control pain throughout the day. It does not cure the disease but manages symptoms.
- The analgesic effect is expected within 30–60 minutes of each dose.
- Diclofenac gel acts locally on the joint to reduce inflammation and pain. Effect is felt within a few days of regular use.
- These medications control symptoms but do not reverse cartilage damage.

Side Effects

Drug	Common Side Effects	Serious / Warn
Paracetamol	Generally well tolerated	Liver toxicity if >4 g/day or with alcohol
Topical Diclofenac	Local skin redness, dryness, itching	Avoid on broken skin or mucous membranes
Oral NSAIDs (if used)	GI upset, nausea, fatigue, ankle edema	GI bleeding, renal impairment, elevated BP

Warnings

- Do NOT stop medications without consulting your doctor, even if you feel better.
- Do NOT exceed the prescribed dose of paracetamol (maximum 4 g per day).
- If you are prescribed a new medication for any other condition, inform the doctor about all current medications.
- Avoid alcohol while taking paracetamol — this significantly increases the risk of liver damage.
- Apply diclofenac gel only to intact skin. Do not cover with occlusive dressings.
- Contact your doctor if you experience unusual bruising, black stools, or significant worsening of pain.

9. NEXT APPOINTMENT

- Review appointment in 4–6 weeks to assess pain control, tolerability, and compliance.
- Instruct patient to keep a pain diary and note any side effects between appointments.
- At each visit: check weight, blood pressure, and ask about GI symptoms.
- If oral NSAIDs are initiated: monitor renal function (creatinine, eGFR) and blood pressure after 2 weeks.

- Reassess need for physiotherapy referral and orthopedic surgery consultation if inadequate response after 3–6 months.

10. CLOSING

To confirm the patient has understood all the information provided, ask her to repeat back the key points:

- What is her diagnosis and what does it mean for her daily life?
- What medications was she prescribed, how to take them, and for how long?
- What lifestyle changes must she make (weight loss, exercise, diet, activity)?
- What side effects to watch for and when to contact the doctor?
- When is her next appointment?

Confirm that the patient has no remaining questions before concluding the consultation.